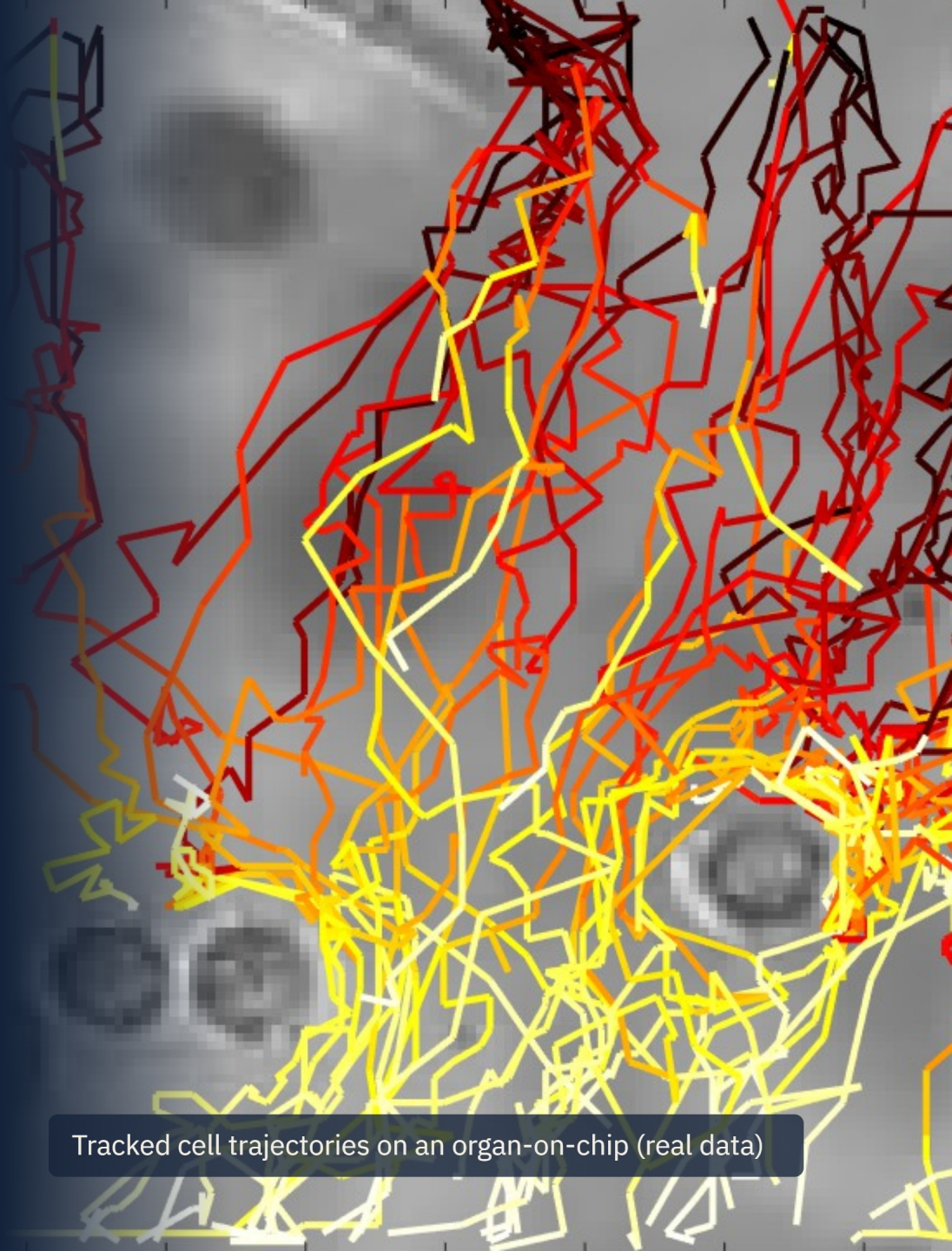
CSMI MASTER 2 · SEMESTER 3 PROJECT

3D Kernel Density Estimation for Cell Tracking on Cancer-on-Chip

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IAC-CNR, Rome · D3 4Health project

elishanchristian.braun@cnr.it · 5 October 2026



Tracked cell trajectories on an organ-on-chip (real data)

CONTEXT

Who we are

IAC-CNR “Mauro Picone”, Rome

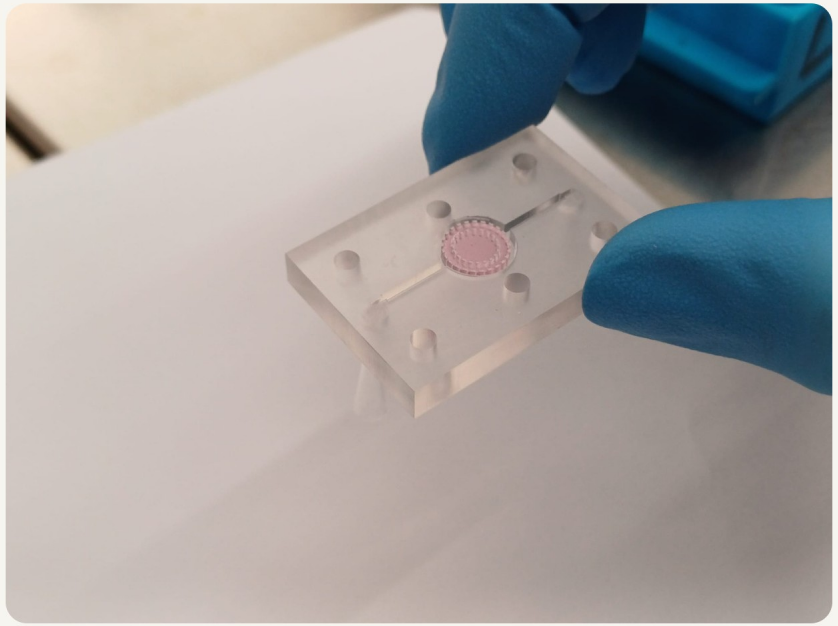
Istituto per le Applicazioni del Calcolo of the National Research Council of Italy: applied mathematics and scientific computing.

D3 4Health

Digital Driven Diagnostics, Prognostics and Therapeutics for Sustainable Healthcare: an Italian PNC/PNRR initiative on digital and biological twins. Our part is the organ-on-chip digital twin.

Collaborators

G. Bretti, S. Bertoluzza, M. Pennacchio · A. De Giorgi, F. Gervaso (CNR Nanotec) · C. Prud’homme (CEMOSIS, Unistra)



Brain cancer-on-chip used in the experiments

CONTEXT

Where the project sits in the pipeline



Goal

Turn organ-on-chip experiments into calibrated, predictive digital models.

Missing link

A reproducible path from microscope images to quantities a PDE can be compared with.

Downstream

Calibration of tumour–astrocyte migration models (A. De Giorgi, CNR Nanotec).

DATA

From microscope images to trajectories

4	23	14	1024^2
channels	z-slices	time frames	pixels per slice

Example acquisition: Zeiss LSM 700 confocal · U87 glioblastoma cells and SVGA astrocytes in Matrigel

How we build them

1. Pre-process the CZI stack (channels × z × t)
2. Segment and detect cells, take 3D centroids
3. Link detections frame to frame into tracks
4. Export one table: id, t, x, y, z, cell type, ROI, track quality

Where it gets hard

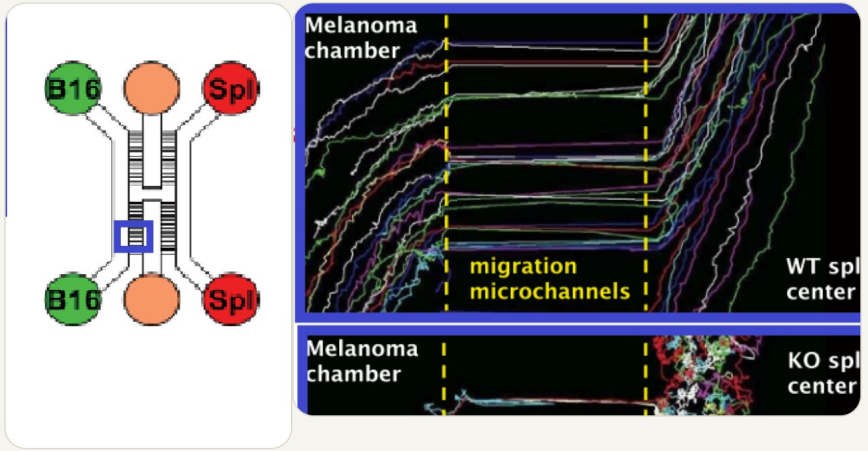
- Only 14 frames: large jumps, ambiguous links
- Coarse z-step: one cell split, or two merged
- Cells leave the field of view or focus: broken tracks
- Touching cells in dense co-culture

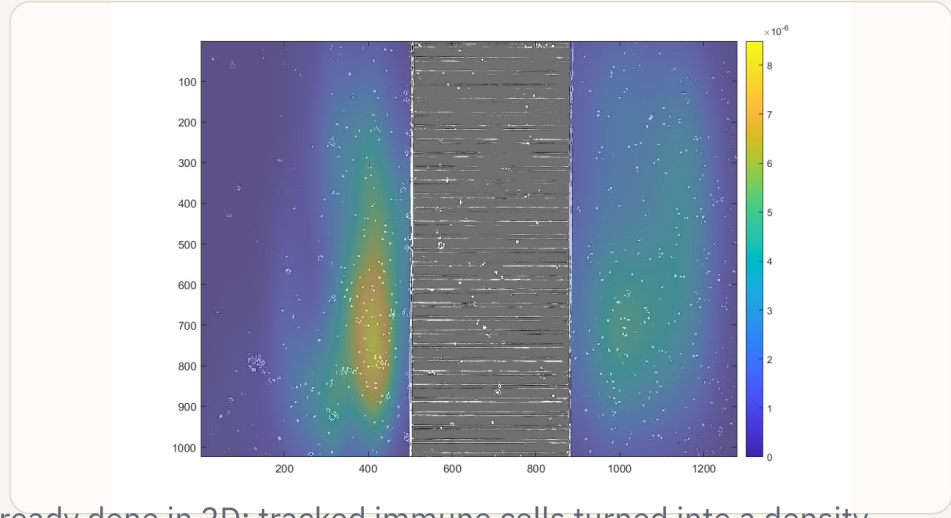
METHOD

Micro to macro: kernel density estimation

$$\hat{u}_p(\mathbf{x}, t_k) = \frac{1}{N_p(t_k)} \sum_{i=1}^{N_p(t_k)} K_{\mathbf{H}}\left(\mathbf{x} - \mathbf{X}_i^p(t_k)\right)$$

- Tracked cells are particles; the PDE model works with densities. KDE turns positions into a smooth field, one per population p and frame k.
- The bandwidth matrix H controls the result: too small gives noisy peaks, too large hides structure.
- The normalisation must match the model: probability, cell number or physical density.

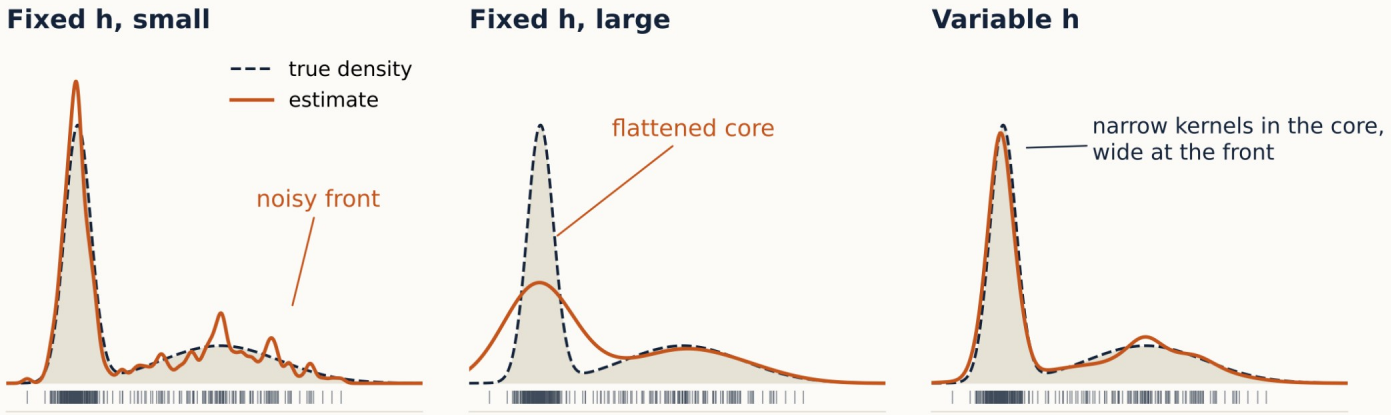




Already done in 2D: tracked immune cells turned into a density field on the chip (immune–tumour cross-talk experiment)

METHOD

Variable kernel and bandwidth: one h does not fit all



$$\hat{f}(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} K_{\mathbf{H}(\mathcal{D})}\big(\mathbf{x} - \mathbf{X}_i\big), \qquad \mathcal{D} = \{\mathbf{X}_1, \ldots, \mathbf{X}_n\}$$

$$K_{\mathbf{H}(\mathcal{D})}(\mathbf{u}) = \big|\mathbf{H}(\mathcal{D})\big|^{-1/2}\, K\big(\mathbf{H}(\mathcal{D})^{-1/2}\,\mathbf{u}\big)$$

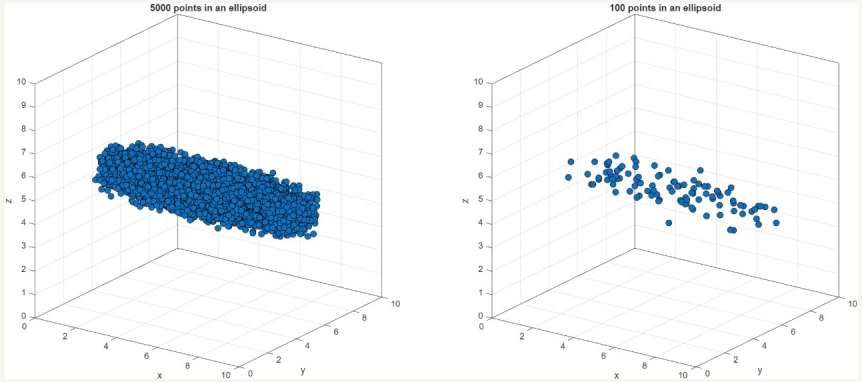
Cells crowd in the tumour core and thin out at the invasion front. The bandwidth matrix **H(D)** is set by the data and its properties in a biological and physical framework.

Synthetic 1D example, 300 samples · Abramson’s square-root law (1982)

METHOD

Why 3D is harder than 2D

100 points **5000 points**



Points sampled in a 3D ellipsoid: radial profile vs the target (1 – r²)^p

Today: 2D slices, projections or thin slabs.
Goal: fully 3D.

Sparser data

$$\mathrm{MISE}^* \propto n^{-4/(d+4)}$$

Same accuracy needs far more cells than in 2D.

Anisotropic voxels

23 slices vs 1024 × 1024 pixels: H needs its full 3 × 3 form, six entries.

Boundaries

Chamber walls, glass bottom, gel edges: kernels leak mass out of the domain.

Cost and checks

Dense grids per frame and population; no ground truth, so synthetic tests.

METHOD

Bandwidth in 3D: a 3 × 3 matrix

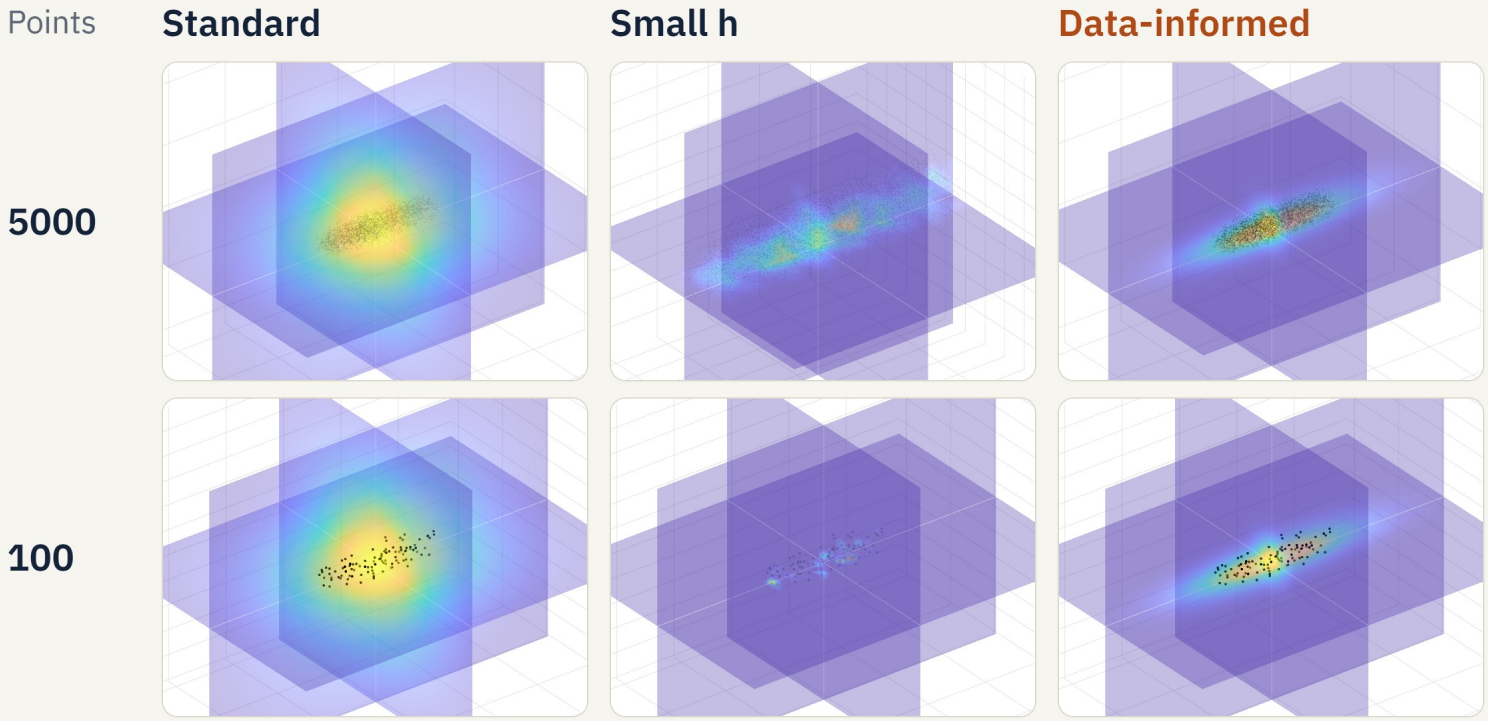
$$\mathbf{H} = \begin{pmatrix} h_{xx} & h_{xy} & h_{xz} \\ h_{xy} & h_{yy} & h_{yz} \\ h_{xz} & h_{yz} & h_{zz} \end{pmatrix}$$

Symmetric positive definite: 6 entries to choose

Standard: H = h²I, one h from an error criterion

Small: H = h²I with a very small h

Data-informed: full H, shaped by the data

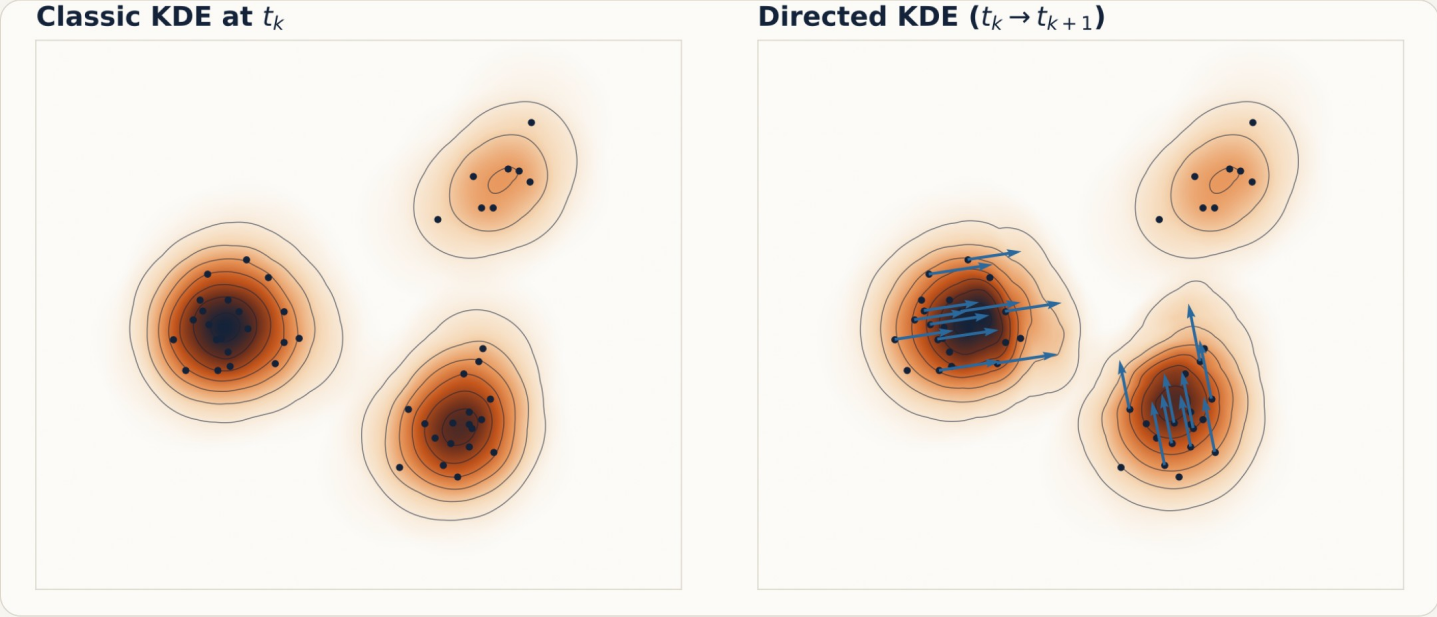


Even with 100 points the data-informed H recovers the elongated shape; a scalar h blurs it or breaks it into spots.

Synthetic test: points sampled in an ellipsoid · colour scales differ between panels

METHOD

Adding time: directed kernel density estimation



Synthetic 2D illustration with tilted cone kernels

Classic KDE

One map per frame; where cells are heading is lost.

Directed KDE

Each kernel tilts toward the cell's next position: one map shows position and trend.

Open, and in scope

3D, full trajectories instead of two time points, speed in the kernel shape.

THE PROJECT

Objectives in three tiers

CORE · MUST REACH

- 3D KDE with a full bandwidth matrix and data-driven selection
- Validation on synthetic 3D data with a known density: error vs n and H
- Applied to one real confocal dataset, viewable in ParaView

EXTENSION

- Variable (adaptive) bandwidth
- Boundary correction for the chip geometry
- Fast evaluation: HCP, GPU

STRETCH

- Directed, time-aware KDE in 3D built from trajectories
- Density fields passed to the PDE model for calibration

Success measured by error on synthetic benchmarks, runtime and quality on real data

THE PROJECT

Where we stand and where it leads

Done in 2D: workflow validated on U87 cells

Open: few SVGA cells → sparse-sampling noise

1 · THIS PROJECT

3D bandwidth matrix

Full SPD H; anisotropy from persistence, velocity covariance, chip geometry

2 · THIS PROJECT (POSSIBLE)

3D ↔ 2D projections

Infer 3D densities from 2D projections or a few focal planes

3 · PHD

Intensity to density

Fluorescence or contrast → equivalent KDE field where cells overlap

4 · THIS PROJECT

Adaptive KDE

Adaptive bandiwidth and kernel for sparse data

5 · THIS PROJECT (POSSIBLE)

Micro–macro bridge

Synthetic trajectories from densities or PDE solutions

6 · PHD

Chemoattractant maps

Inverse reaction–diffusion problem for hidden chemical fields, in 3D

Bandwidth benchmarks: Wand & Jones (1995); Duong & Hazelton (2003, 2005)

THE PROJECT

Mathematics and computing

Mathematics

- Non-parametric statistics: KDE, bias–variance trade-off, MISE and AMISE
- Bandwidth selection as an optimisation problem
- Data-Analysis
- Numerical analysis: convergence studies
- Numerical Methods for PDE-ODE, Inverse problems and calibration, PINN

Computing

- Python (NumPy, SciPy) to prototype; C++ for the heavy kernels
- Starting points: our 2D KDE code, the image-to-trajectory package (feelpp/ooc)
- ParaView and VTK for 3D results
- Git and GitHub, tests, reproducible runs

FIELDS

Data and machine learning | Modelling, simulation, optimisation and control | High-performance computing | Signal and image processing

THE PROJECT

Profile, deliverables and next steps

Student profile

- Interest and enthusiasm in scientific computing
- Matlab, Python; C++ is a plus
- No biology background needed
- Team size adaptable: two students can split data and methods

Deliverables

- Documented, tested code in the project repository
- Report: synthetic validation and real-data results
- Final oral presentation

After the project

- **PhD:** yes
- **Internship:** possible, to be discussed
- Supervision from IAC-CNR Rome

Questions or interest? **elishanchristian.braun@cnr.it**